

Shiqi (Stella) Chen

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Personal Profile

System and software engineer specializing in backend systems and platform infrastructure for healthcare applications serving 2M+ patients across 900+ hospitals and clinics. Broad technical foundation spanning distributed systems, machine learning, cryptography and security.

Experience

PocketHealth

Toronto, Canada

Internship - Software Engineer (PEY)

May 2024 - Aug 2025

- Developed Prometheus-based monitoring and internal alerting to track backend service health, traffic patterns, and operational reliability across distributed systems.
- Automated monthly Azure storage utilization reporting with cron-based pipelines, improving infrastructure cost visibility and reducing manual tracking for the finance team.
- Implemented JWT authentication to secure service-to-service communication across a distributed backend architecture.
- Designed and optimized dynamic GitLab CI/CD pipelines, modularizing infrastructure workflows and reducing deployment overhead.
- Maintained Kubernetes-based backend infrastructure on Azure, supporting containerized services, ingress controllers, deployment automation, and internal platform operations.
- Supported security hardening for Angular-based frontend applications by addressing browser-side restrictions and security scan findings.
- Collaborated with platform and backend engineers to maintain deployment pipelines, troubleshoot infrastructure issues, and improve reliability of containerized services in QA environment.

Kua.ai

Zhuhai, China

Internship - Software Engineer

June 2023 - Aug 2023

- Reverse-engineered Midjourney's interface to build a seamless integration for Kua.ai, bridging the gap between raw generative AI capabilities and the high precision requirements for cross-border e-commerce product listing.
- Collaborated across the development team to overcome technical constraints and deliver a reliable end-user experience.

University of Toronto

Toronto, Canada

Work Study - Research Assistant (Supervisor: Prof. Qun Wang)

Jan 2025 - Apr 2025

- Built an N-Vortex simulator for analyzing vortex dynamics through simulation-based experimentation, testing, and iterative validation.

Work Study - Lecture Notes Assistant (Supervisor: Prof. Lisa Jeffrey)

Jan 2024 - Apr 2024

- Prepared and refined lecture notes in elementary Lie theory.

Teaching Assistant - MAT135, MAT233, MAT102, MAT224, MAT223, MAT136, CSC236, CSC263, CSC347, CSC427

Jan 2023 - Present

- Led tutorials for math and CS courses. Coordinated tutorial logistics and developed course assessments to support course delivery.

Education

University of Toronto

Toronto, Canada

Honours Bachelor of Science with High Distinction (CGPA: 3.95/4.00)

Sept 2021 - June 2026

- Computer Science Specialist; Information Security Specialist; Mathematical Science Minor.
- Dean's List Scholar: 2022 Summer, 2023 Winter, 2026 Winter.

International Department, The Affiliated High School of South China Normal University

Guangzhou, China

High school

Aug 2018 - June 2021

Selected Projects

Medical Image Classification

Sept 2025 - Dec 2025

- Built a semi-supervised chest X-ray classification pipeline on NIH ChestX-ray14 using MoCo pretraining and Mean Teacher pseudo-labeling under limited labeled data.
- Designed weak/strong augmentation workflows and tuned teacher-student consistency objectives to improve pseudo-label quality and classification performance.
- Optimized PyTorch training on NVIDIA H100 GPUs using torch.compile, TensorFloat-32, channels-last memory format, dataloader tuning, and local data caching.
- Reduced per-epoch training time from approximately 12 minutes to 3 minutes and improved model AUC from 57% to 63% after semi-supervised training.

Formal Verification of PBFT Consensus

Nov 2025 - Dec 2025

- Modeled PBFT in TLA+ and used formal verification to analyze safety and fault tolerance in distributed consensus.
- Examined smart contract semantics and their dependence on protocol-level guarantees for secure decentralized execution.

Skills

Languages

Go, Python, Java, Kotlin, C/C++, SQL, TypeScript, UNIX/Shell, HTML/CSS, Solidity

Infrastructure & Tools

Skills, Git, Docker, Kubernetes, Terraform, GitLab CI, Prometheus, Microsoft Azure, 四X

Areas

Distributed Systems, Cryptography, Blockchain, Machine Learning, Computer Security, Infrastructure Automation